

LAKSHIT VERMA

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TECHNICAL EXPERIENCE

Student Software Developer, FLARE Mandiant

May 2026 – August 2026

Google Summer of Code

Remote

- Extending FLOSS with QS branch for 25+ OSS libs, 100K+ strings, context-aware tagging and section rendering.
- Implementing PE/ELF/Mach-O parsing with section-aware extraction for memory-dumped and standard binaries.
- Building React/Vite/Bun GUI handling 100K+ strings via virtualization with Fuse.js fuzzy search achieving <100ms latency.
- Designing CI/CD pipeline with GitHub Actions/vcpkg for weekly builds of 25+ OSS libs, maintaining 17+ databases.

Undergraduate Researcher

September 2025 – Present

Dr. T.M.A. Pai Endowment Chair

Manipal, Karnataka, India · Hybrid

- Working under Senior Professor Dr. Manohara Pai and Schneider Electric on AI-enabled security analysis and test case generation.
- Built a novel framework via LLMs to automatically parse CVE, CWE and MITRE ATT&CK databases across 335K+ records.
- Automated binary vulnerability analysis using Radare with exploit generation and sandboxed verification in controlled environments.

Student Software Developer, Dhvani RIS

May 2025 – August 2025

Code for GoodTech DMP

Remote

- Developed a FastAPI-based backend parser to enable bulk conversion of Excel-based surveys to mForm compatible JSON files.
- Built an Angular frontend from scratch for real-time async Excel uploads, live previews and inline error reporting.
- Integrated frontend with REST APIs and MongoDB for seamless form lifecycle management.
- Achieved batch processing speeds of 2.16 seconds for 9 forms, supporting forms with up to 400+ questions per upload.
- Created comprehensive testing and CI/CD pipelines, validating JSON and implementing error handling in frontend and backend.

Digital Forensics Head

January 2024 – Present

Team Cryptonite — Major Student Project

Manipal, Karnataka, India · On-site

- Participated in 200+ Capture The Flag (CTF) competitions, achieving #3 national ranking in 2024 and 2025 on ctftime.org.
- Designed and developed 5+ challenges for niteCTF 2024 and 2025, focusing on real-world digital forensics scenarios.
- Trained 11 Junior Members in digital forensics techniques, tools, and methodologies, enhancing the team's overall skill set.
- Created the "DFIR Gita" repository with 25+ stars, a comprehensive incident response training resource for future team members.
- Conducted a guest lecture on a DFIR workflow for Master's students at MIT, sharing insights through a real-world case study.

ACHIEVEMENTS

- Awarded **#1 Position** and **2,00,000 INR** in the ISEA-ISAP 2026 Hackathon organized by IIT Madras.
- Awarded **#1 Position** and **1,00,000 INR** in the Smart India Hackathon 2024, held at IIT Jammu.
- Awarded **#1 Position** and **25,000 INR** in the GITxIITB CTF by KLS GIT, Belagavi & IIT Bombay Trust Lab.
- Awarded **#1 Position** and **15,000 INR** in the KJSSE CTF by KJ Somaiya College of Engineering, Mumbai.
- Awarded **#2 Position** and **10,000 INR** in the SoftLaunch Hackathon by the MAHE Innovation Centre.

PROJECTS

NiteWatch

Python, PyQt, Btrfs, DFIR, Operating Systems

- DFIR application for Btrfs/XFS filesystems restoring deleted files with complete metadata and superblock parsing.
- Parses B+ trees and inode records with PyQt GUI for visualizing reconstructed filesystem structures.

Astraeus — [Source Code](#)

Python, PyTorch, Three.js, XGBoost

- An ensemble ML application pipeline combined with LSTM/GRU networks for high-accuracy trajectory and collision assessment.
- Provides real-time ingestion and preprocessing of Celestrak TLE data with feature engineering on orbital elements.
- Implements end-to-end system delivering 100ms inference latency, continuous risk monitoring, and 3D viz. of orbital dynamics.

MFT Parser — [Source Code](#)

C++, CMake

- Cross-platform C++ NTFS forensic parser and CLI for MFT with metadata extraction and deleted file recovery.
- Implements resident and non-resident cluster mapping for forensic workflows on macOS, Linux, and Windows.
- Uses portable CMake-based build system with recursive artifact detection for analysis on all platforms.

EDUCATION

Manipal Institute of Technology, Manipal, Karnataka

July 2023 – August 2027

Bachelor of Technology — Electrical Engineering (Minor in Computing)

Currently in VIth Semester

Languages

Python, C/C++, TypeScript, Bash, HTML/CSS, L^AT_EX, SQL

Frameworks & Libraries

Angular, FastAPI, Material UI, React, Next.js, PyQt, Node.js, Express.js, Tailwind CSS

Technologies

MongoDB, REST APIs, JSON, XLSForm, Git, Docker, Make/CMake

Cybersecurity

IDA Pro, Ghidra, Wireshark, Volatility, GDB, EnCase, Sleuthkit, FTK Imager, Btrfs